

Question1: Answer the following:

(7 M)

Let $x = 0.54617$ and $y = 0.54601$.

Use **four-digit arithmetic** to approximate $x - y$ and determine the absolute and relative errors using rounding of numbers.

Solution:

The exact value r of $x - y$ is $r = 0.00016$.

1

Rounding x and y to four-digit gives: $\tilde{x} = 0.5462$ and $\tilde{y} = 0.5460$

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Then: $\tilde{r} = \tilde{x} - \tilde{y} = 0.5462 - 0.5460 = 0.0002$ is the four-digit approximation to r and:

1

Absolute error = | Exact value - Approximate value | = | 0.00016 - 0.0002 | = 0.00004

2

Relative error = | Exact value - Approximate value | / | Exact value | =

$$| 0.00004 | / | 0.00016 | = 0.25$$

2

Question 2: Given the following polynomial:

(8 M)

$$P(x) = 3x^3 - 4x^2 - 12.$$

a) Use Horner's algorithm to evaluate $P(-2)$.

b) Using the complete Horner's algorithm, find the Taylor expansion of $P(x)$ about the point $r = 3$.

a) 3

$$\begin{array}{r} 3 \quad -4 \quad 0 \quad -12 \\ -2 \quad \quad -6 \quad 20 \quad -40 \\ \hline 3 \quad -10 \quad 20 \quad -52 \end{array}$$

2

Thus, we obtain $P(-2) = -52$.

1

b) 5

$$\begin{array}{r} 3 \quad -4 \quad 0 \quad -12 \\ 3 \quad \quad 9 \quad 15 \quad 45 \\ \hline 3 \quad 5 \quad 15 \quad 33 \\ \quad 9 \quad 42 \\ \hline 3 \quad 14 \quad 57 \\ \quad 9 \\ \hline 3 \quad 23 \end{array}$$

1

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Thus, we can write:

$$P(x) = 3(x - 3)^3 + 23(x - 3)^2 + 57(x - 3) + 33$$

2

Question 2: Floating-point representation**(7 M)**

Determine the single-precision machine representation in a 64-bit word-length computer of the following decimal number (-45.21875) .

Solution:

1. Converting the integer part to binary, we have:

$$(45)_{10} = (55)_8 = (101101)_2. \quad 1$$

2. Converting the fractional part to binary, we have:

$$(.21875)_{10} = (.16)_8 = (.001110)_2. \quad 1$$

3. $(45.21875)_{10} = (101101.001110)_2 = (1.01101001110)_2 \times 2^5$ is the corresponding one-plus form in base 2, and $(.01101001110)_2$ is the stored mantissa. 1

4. Next the exponent is $(5)_{10}$, then $c - 127 = 5$, we immediately see that $(132)_{10} = (204)_8 = (1000\ 0100)_2$ is the stored exponent. 2

Thus, the single-precision machine representation of (-45.21875) is:

$$[110000100011010011100000000 \dots 0000]_2 \quad 1$$

$$[1100\ 0010\ 0011\ 0100\ 1110\ 0000 \dots 0000]_2 = [C234E000]_{16} \quad 1$$

Question 3: Given the following function:**(8 M)**

$$f(x) = x^3 - 9x + 1$$

- a) Prove that the function $f(x)$ has exactly one real root α in the interval $[2, 4]$. 2
 b) Find a root of the function $f(x)$ on the interval $[2, 4]$ by using the bisection method. Terminate the iterations if they exceed $n_{max} = 2$ or if the error bound is less than 0.5×10^{-6} .

Solution:

a.

$$u = f(2) = -9$$

$$v = f(4) = 38$$

Since, $u \cdot v < 0$ then the function f has exactly one real root α in the interval $[2, 4]$. 2

b.

- 1) $n = 0$ and $x_0 = (2 + 4)/2 = 3$, $w = f(x_0) = f(3) = 1$

$u \cdot w < 0$ then: $a = a$, $b = c$ and the interval $[2, 3]$. Error = 1. 2

- 2) $n = 1$ and $x_1 = (2 + 3)/2 = 2.5$, $w = f(x_1) = f(2.5) = -5.875$

$u \cdot w > 0$ then: $a = c$, $b = b$ and the interval $[2.5, 3]$. Error = 0.5. 2

- 3) $n = 2$ and $x_2 = (2.5 + 3)/2 = 2.75$, $w = f(x_2) = f(2.75) = -2.9531$

$u \cdot w > 0$ then: $a = c$, $b = b$ and the interval $[2.75, 3]$. Error = 0.25. 2

n	x_n	$f(x_n)$	Error
0	3	1	1
1	2.5	-5.875	0.5
2	2.75	-2.9531	0.25

Not: Skipping any step will result in losing a grade.